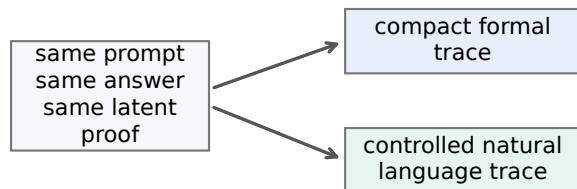


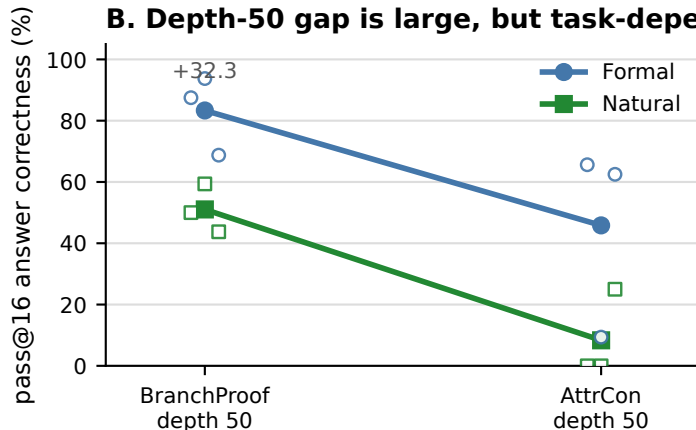
# Formal traces improve extrapolative answers, but proof validity remains a separate target

## A. Same proof, different trace language

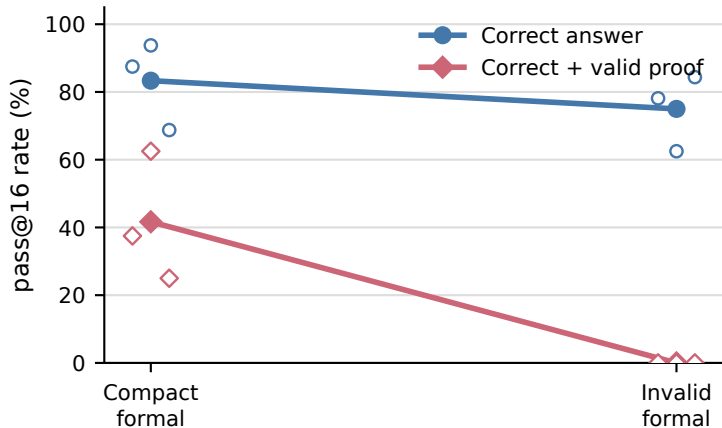


Only the trace substrate changes.

## B. Depth-50 gap is large, but task-dependent



## C. Correct answers need not be valid proofs



## D. Samples expose success and truncation

### Formal depth-50 success

```
gold=olive | answer=olive | valid=1  
<formal> <constants> a = a b = b c = c  
d = d e = e f = f g = g h = h i = i j =  
j k = k l = l ...
```

### Natural depth-50 failure

```
gold=juniper | answer=missing | valid=0  
<think> <premises> a is orchid. a is  
east. If a is south and a is orchid,  
then b is cedar. I ...
```

Excerpts come from sample\_generation\_snippets.csv. Metrics use 16 samples per prompt.